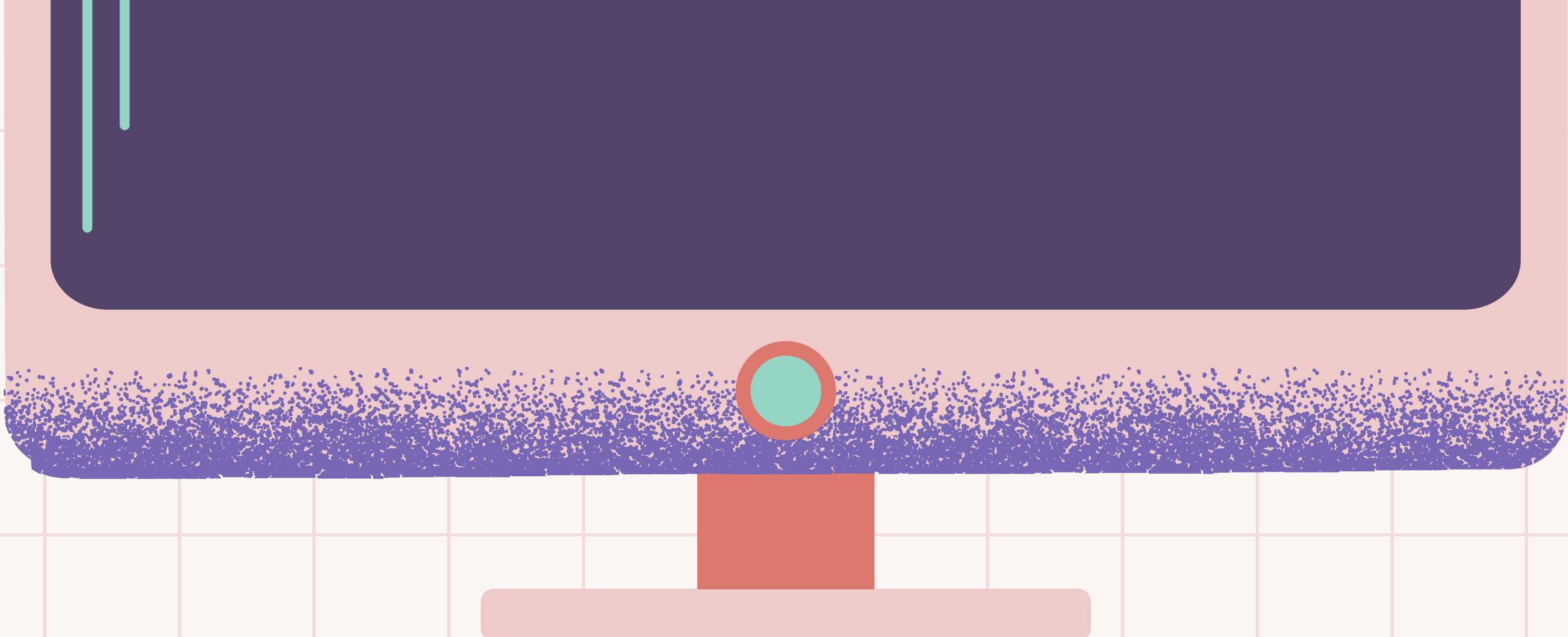




The graphic depicts a stylized device. At the top is a dark purple rounded rectangle representing a screen, with two thin teal vertical lines on its left side. Below the screen is a light pink horizontal bar. Underneath this bar is a dense horizontal band of small blue dots. Centered on this band is a teal circle with a red outline. Below the dots is a solid red rectangle, which sits on a wider, light pink horizontal base.

INTERCONEXIÓN

de Dispositivos

INTRODUCCIÓN

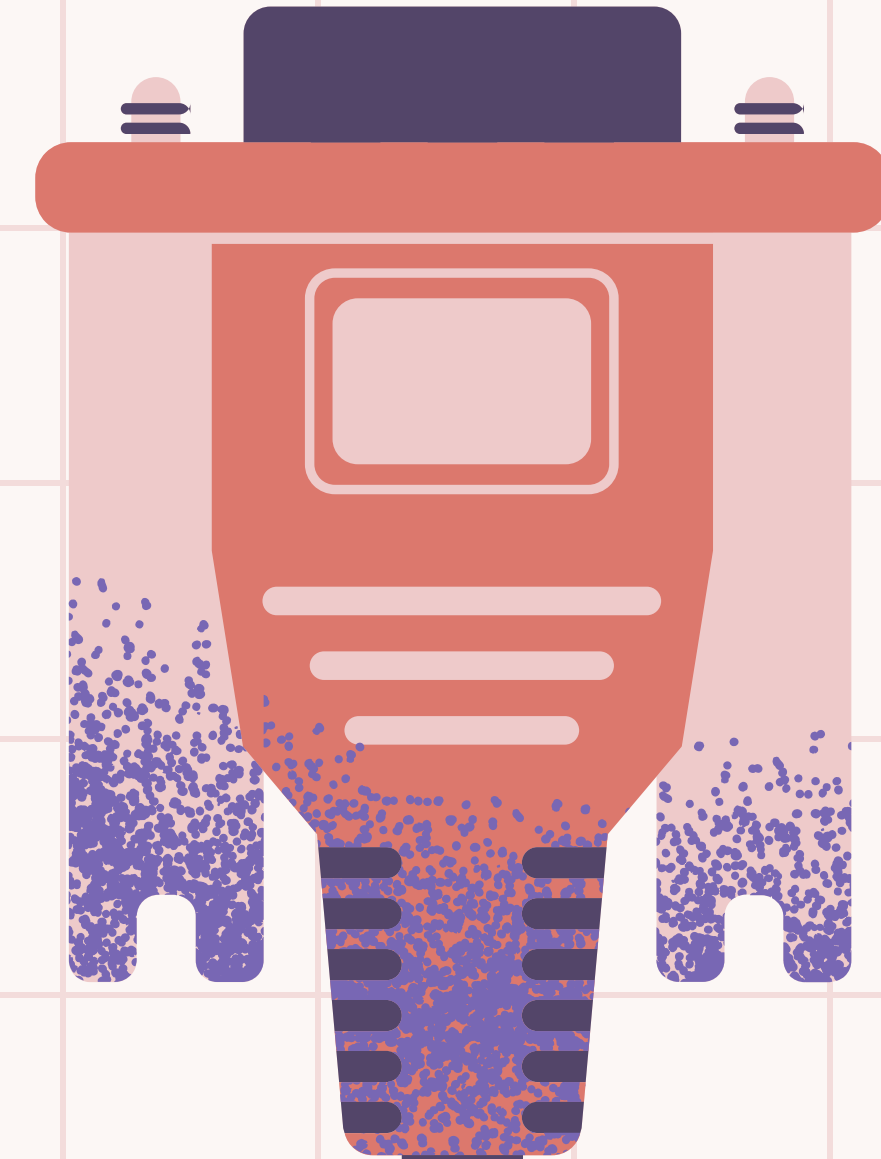
- ICPC - International Collegiate Programming Contest
- Sede del evento - Tecnológico de Monterrey
- Requisitos para la red
 - Sobrecarga
 - Acceso a plataformas y BOCA
 - Comunicación entre organizadores



ICPC
INTERNATIONAL
PROGRAMMING
CONTEST SEASON



2024/2025





¿QUÉ SE DEBE HACER?

Sistema a diseñar



- Integrar una infraestructura en el campus
- Evitar choques con las IP
- Garantizar conectividad estable
- No caerse cuando hay una gran cantidad de usuarios

OBJETIVOS

01

Analizar
requerimientos
de red

02

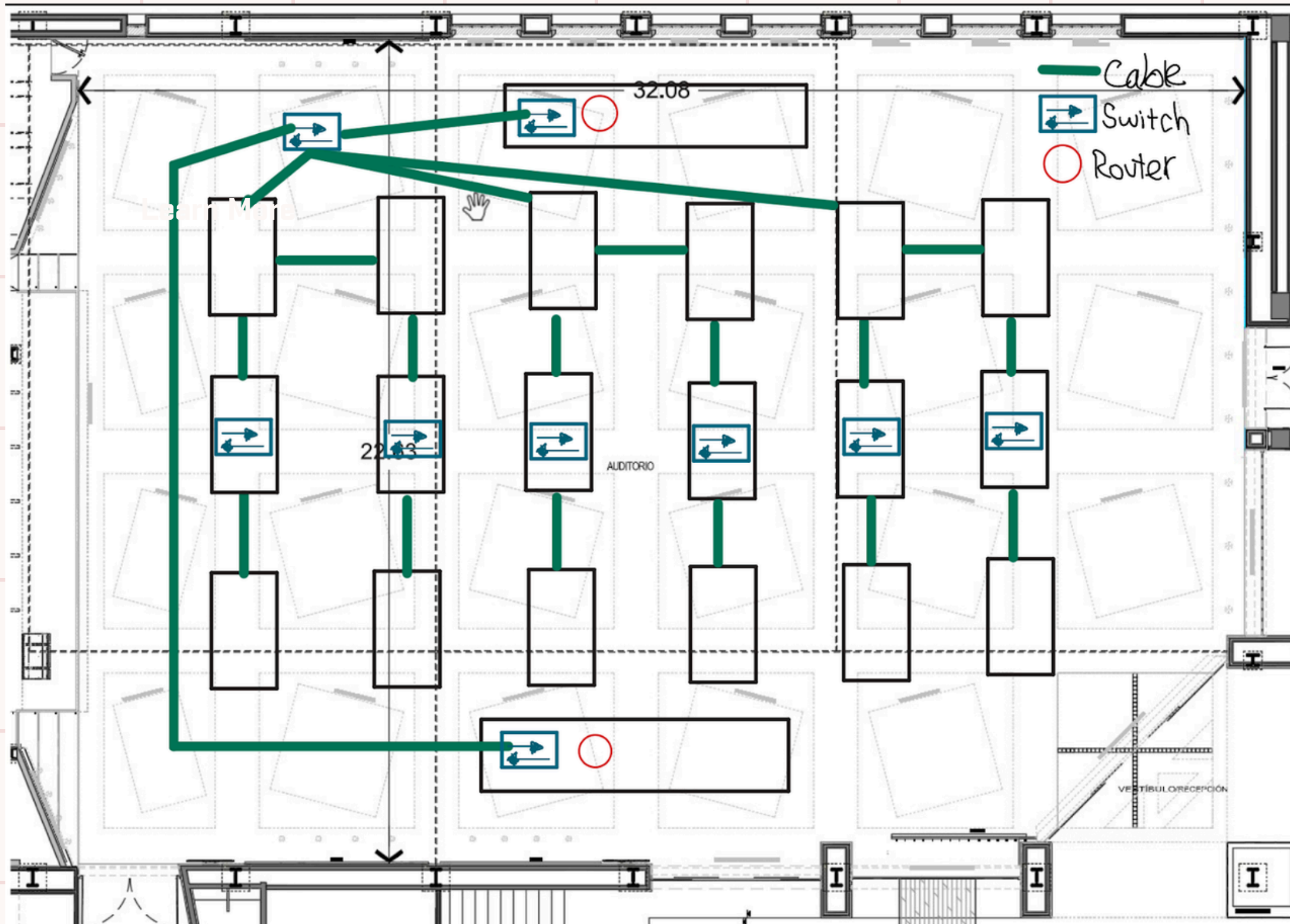
Diseñar
arquitectura

03

Implementar y
validar
conectividad



DISEÑO DE RED



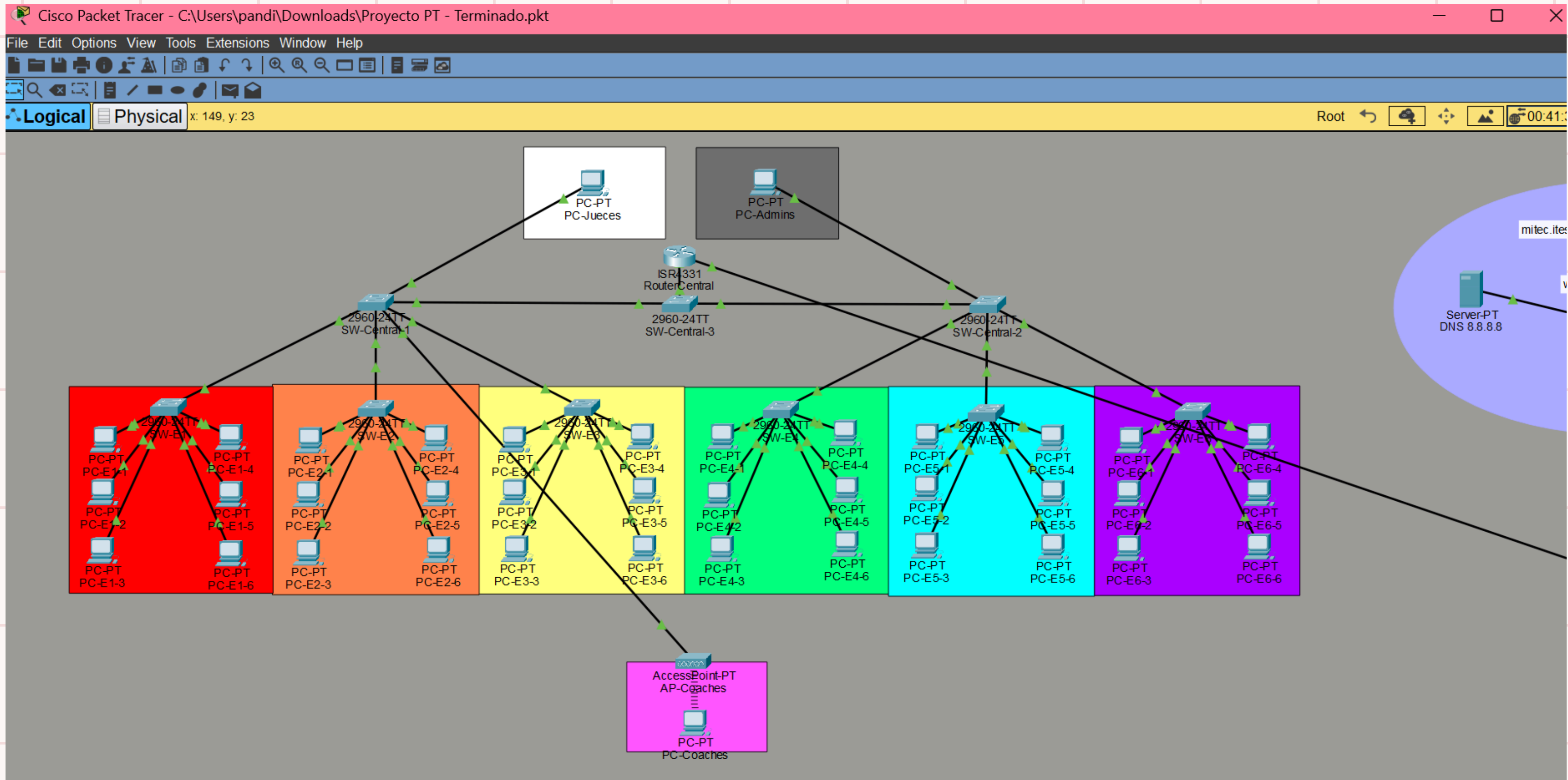
Mínima capacidad de 204 personas

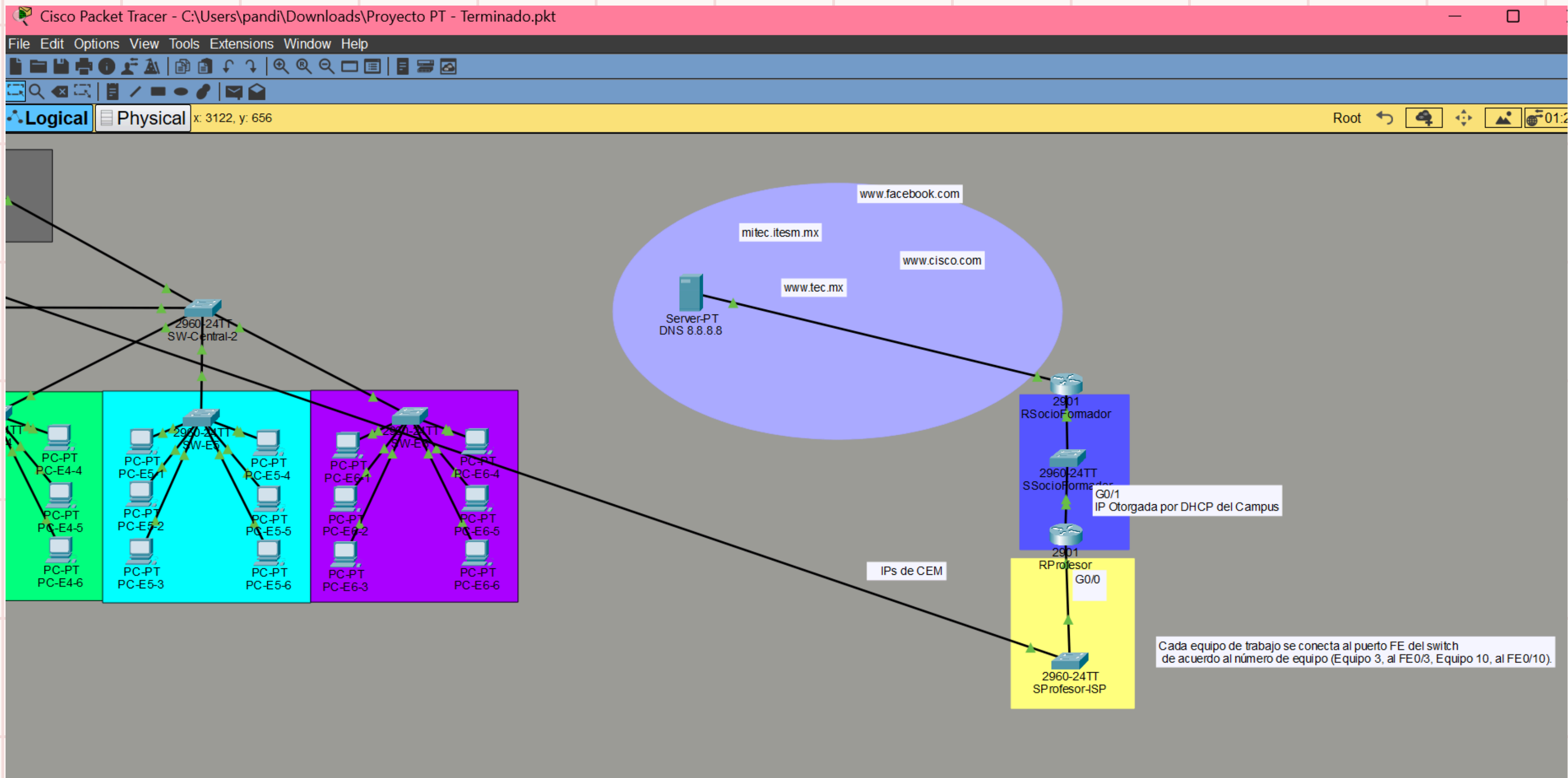
- 6 equipos (30 personas)
- 10 jueces
- 2 administradores
- 12 entrenadores

Infraestructura

- Switches
- Routers
- Access Point
- PCs

DISEÑO DE RED





DISEÑO CON SALIDA INTERNET

TABLA DE SUBREDES

Segmento	Hosts requeridos	Prefijo de red	Máscara en notación punto decimal	Bloque asignado de direcciones IP	Primera dirección IP válida del bloque	Última dirección IP válida del bloque
Equipo 1	39/64	26	255.255.255.192	172.20.24.0 - 172.20.24.63	172.20.24.1	172.20.24.62
Equipo 2	39/64	26	255.255.255.192	172.20.24.64 - 172.20.24.127	172.20.24.65	172.20.24.126
Equipo 3	39/64	26	255.255.255.192	172.20.24.128 - 172.20.24.191	172.20.24.129	172.20.24.190
Equipo 4	39/64	26	255.255.255.192	172.20.24.192 - 172.20.24.255	172.20.24.193	172.20.24.254
Equipo 5	39/64	26	255.255.255.192	172.20.25.0 - 172.20.25.63	172.20.25.1	172.20.25.62
Equipo 6	39/64	26	255.255.255.192	172.20.25.64 - 172.20.25.127	172.20.25.65	172.20.25.126
Coaches	16/32	27	255.255.255.224	172.20.25.128 - 172.20.25.159	172.20.25.129	172.20.25.158
Jueces	13/16	28	255.255.255.240	172.20.25.160 - 172.20.25.175	172.20.25.161	172.20.25.174
Admin	13/16	28	255.255.255.240	172.20.25.176 - 172.20.25.191	172.20.25.177	172.20.25.190

COSTOS

No se considera el costo de los elementos de la salida a internet

Equipo	Modelo	Cantidad	Costo (USD)	Costo total (USD)
Router	Cisco ISR-4331	1	\$4,316	\$4,316
Switch de distribución	Cisco 24TT 2960	3	\$1,424	\$4,272
Switch de acceso	Cisco 2960	6	\$960	\$5,760
Access point	AP-PT	1	\$650	\$650
Computadoras (estudiantes)	PC Estándar	180	\$500	\$90,000
Computadoras (jueces)	PC Estándar	10	\$500	\$5000
Computadoras (administradores)	PC Estándar	2	\$500	\$1000
				Costo total: \$110,998

SW-E1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
User Access Verification

Password:

Switch>enable
Password:
Switch#show run
Building configuration...

Current configuration : 1512 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Switch
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/7
 switchport mode trunk
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
```

Switch Equipo 1

CONFIGURACIÓN

```
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
 no ip address
 shutdown
!
!
!
!
!
line con 0
 password 7 0822455D0A16
 login
!
line vty 0 4
 password 7 0822455D0A16
 login
line vty 5 15
 login
!
!
!
!
end

Switch# |
```

PRUEBAS

Equipo 1 a Internet

Equipo 1 a 4

PC-E1-3

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.20.24.198

Pinging 172.20.24.198 with 32 bytes of data:

Reply from 172.20.24.198: bytes=32 time=10ms TTL=127
Reply from 172.20.24.198: bytes=32 time<1ms TTL=127
Reply from 172.20.24.198: bytes=32 time<1ms TTL=127
Reply from 172.20.24.198: bytes=32 time<1ms TTL=127

Ping statistics for 172.20.24.198:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

PC-E1-4

Physical | Config | Desktop | Programming | Attributes

Command Prompt

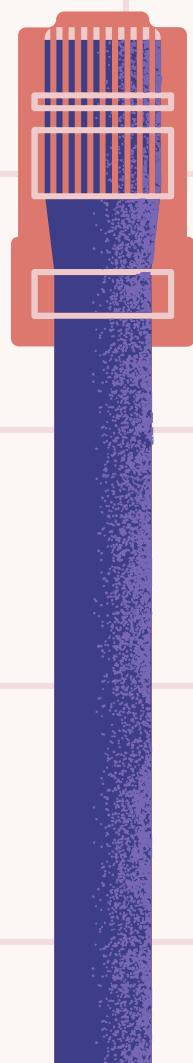
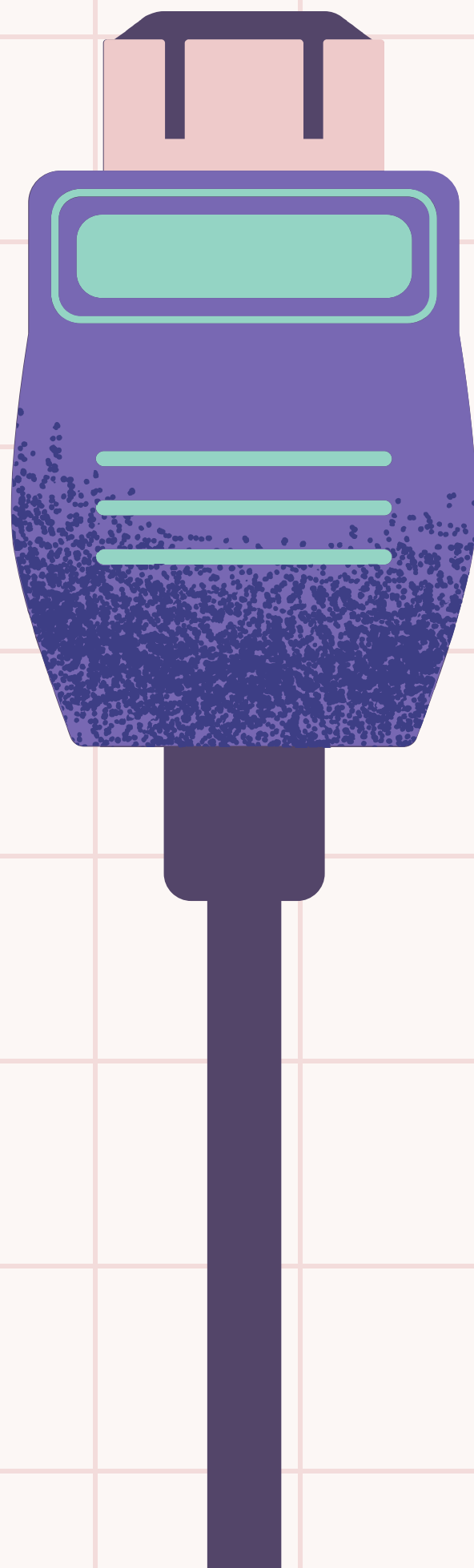
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=21ms TTL=125
Reply from 8.8.8.8: bytes=32 time<1ms TTL=125
Reply from 8.8.8.8: bytes=32 time<1ms TTL=125
Reply from 8.8.8.8: bytes=32 time<1ms TTL=125

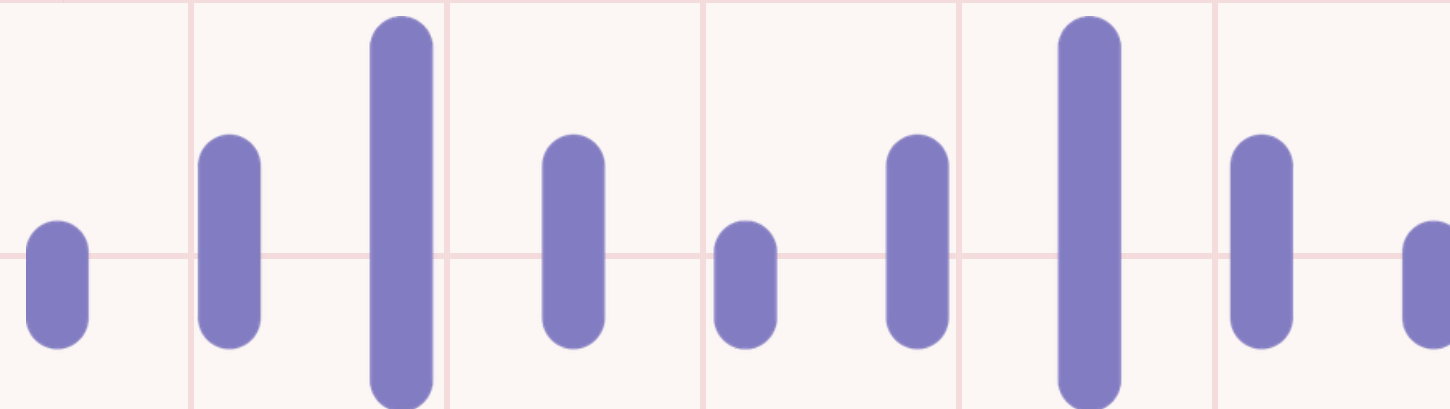
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 21ms, Average = 5ms

C:\>
```



EVALUACIÓN Y RESULTADOS

- Logramos una red funcional
- Segmentación eficiente
- Conexiones estables con la red del campus

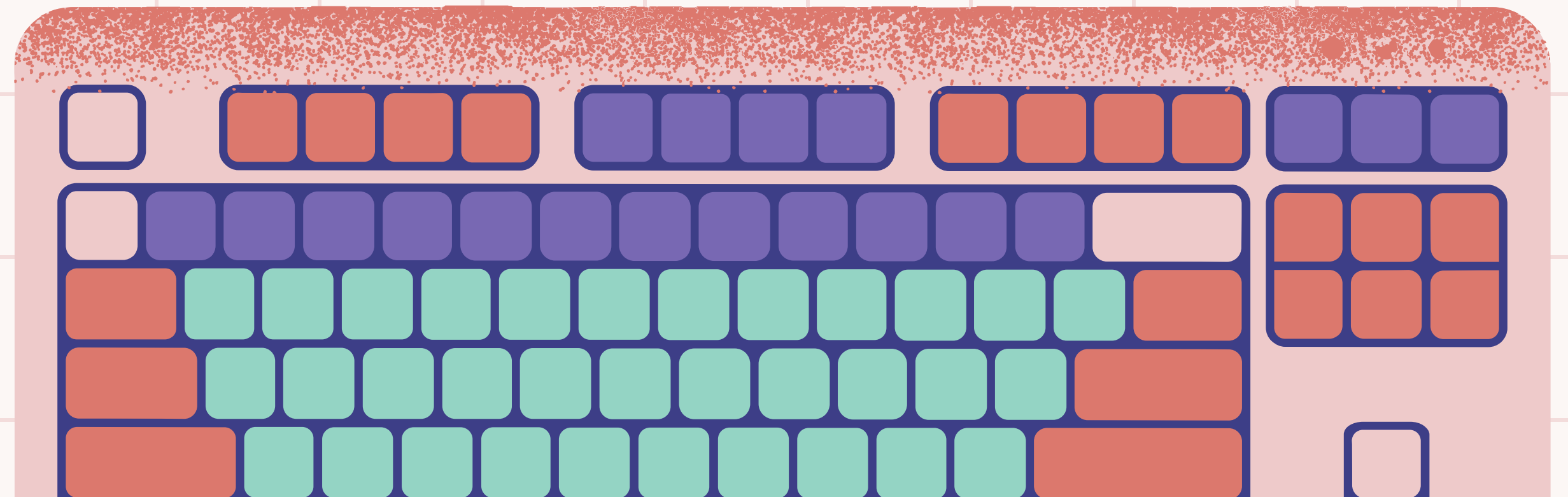


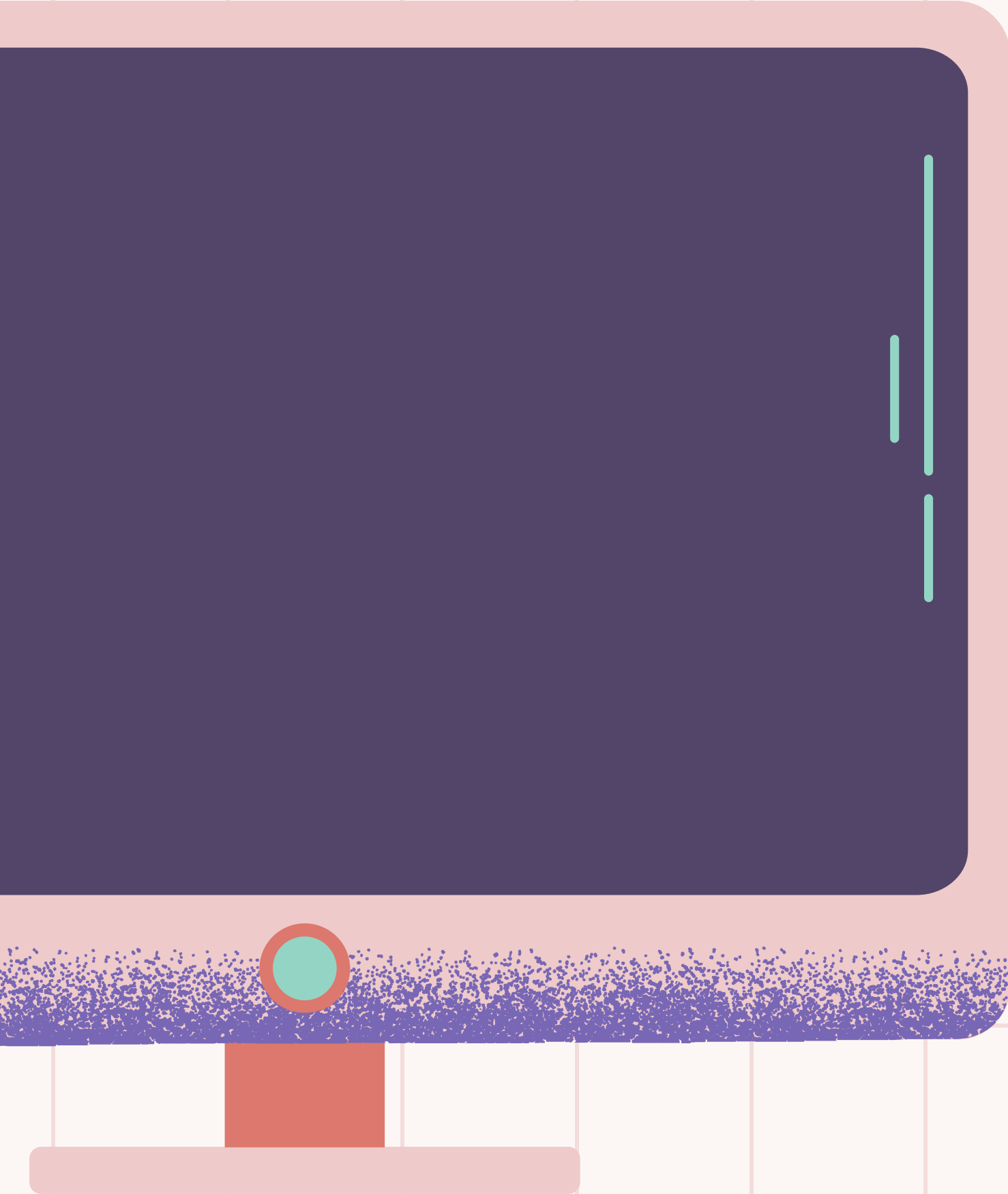
CISCO



PROBLEMAS

- Conectar la salida a internet
- Re-hacer la propuesta para lograr conectar todo
- Sobrecarga del switch central 1





CONCLUSIÓN Y FUTURO

- Red que cumple los requisitos
- Eficiente
- Simulación que cumple las pruebas
- Hacerlo físico



GRACIAS

